

Digital Logic Design

CSE - 260

Assignment 04

Summer 2025

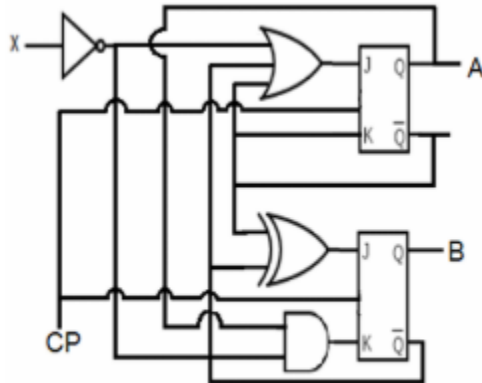
Note that:

- 1. You will have to submit the hard copy, show all steps in all questions*
- 2. Make an cover page which contains you name, student ID and section*
- 3. You do not need to submit the ungraded portion.*

Deadline: 4 September,2025

Graded

1. Draw the state diagram for the given circuit.



Ungraded

1. **Design** a 13 person voter counting system using full and parallel adders. You must use encoder and decoder to build full adders, meaning you can't use full adders directly.
2. Spider-Man is keeping an eye on a big city march. He has four tiny sensors set up around the area. Each sensor can send a "1" if it spots something important:
 - D3 – Rooftop scout: sees leaders forming at the front
 - D2 – Street camera: sees group movement in the middle
 - D1 – Alley microphones: hear shouts or calls for help from side streets
 - D0 – Wearable devices: distress from individual civilians

He needs a device that **picks the most important signal** at any moment, and outputs it as a **2-bit code**. This is done using a 4×2 priority encoder — but with special rules.

The Rules for Priority:

- If there is No "1" signal → output should be don't care.
- Only two or more consecutive "1"s in a row → Treat as a "group event" and The priority goes to the leftmost 1 in that group — the one with the bigger bit position (closer to D3)
- Exactly two "1"s but not consecutive → The priority goes to the rightmost 1 in that group — the one with the smaller bit position (closer to D0)
- Anything else - pick D0

Now, design the circuit diagram of the priority encoder and use K-Map to simplify the function.

3. Count Dracula uses a “Nocturnal Operations Controller” (NOC) with three magical sensors:

Inputs:

- $B = 1$ if many bats are flying nearby (good cover)
- $M = 0$ if the moon is bright (bright light makes Dracula avoid hunting)
- $H = 0$ if humans are detected nearby (potential prey — Dracula prefers 0 here means humans present)

Outputs:

$HUNT = 1 \rightarrow$ Dracula goes out for hunt

$HUNT = 0 \rightarrow$ stays in castle

Rules for Hunt:

1 or 1 and 0

1 and 1 and 1

Hunt = 1 ; If prey is present along with bats flying nearby or the moon is dark.

Hunt = 1; if no prey is present but the Moon is dark and bats are flying nearby.

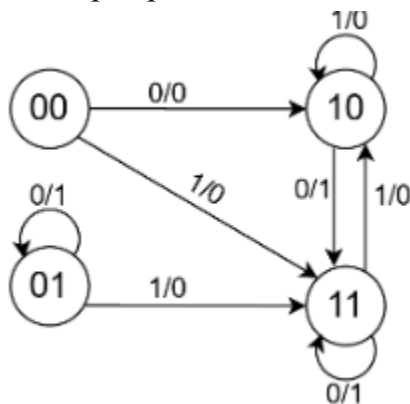
Hunt = 0; Otherwise.

Now, help Dracula by designing this device for him.

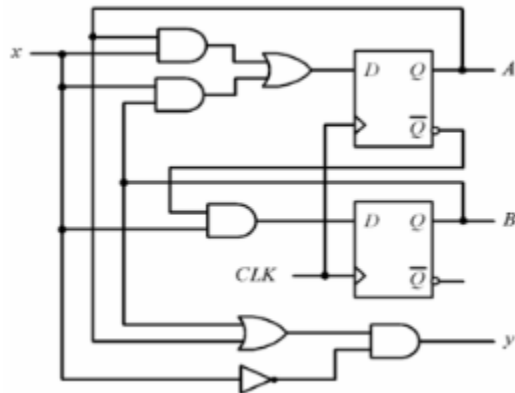
a) Design this “NOC” device using 2x4 decoder(s).

b) Design this “NOC” device using a 4:1 Mux.

4. Design a 3 bit NAND gate using Decoder
5. Design a 2 bit parallel adder using Decoder as internal circuit
- 6.
7. Design a D FF using SR FF.
8. Design a T FF using JK FF.
9. Design a D FF using JK FF
10. Given the state diagram as follows, get the sequential circuit using
- SR flipflop.
 - JK flipflop.



11. Draw the state diagram for the given circuit.



12. Implement the following counter using T flip flop

13. CSE110-> CSE111-> CSE220-> CSE221-> CSE331-> CSE221-> CSE321->
CSE110

14. 3->4->6->10->12->13->15->3

- Implement the given counter using JK flip-flop.
- Implement the given counter using T flip-flop.

NB: For states not given in the question, please move to the initial state as per the question.

15. Implement a 4 bit up counter using JK flip-flop.

16. Implement 4 bit down counter using JK flip-flop

17. Implement the following counter using T FF:

Green->Orange->Yellow->Red->Yellow->Orange->Yellow->Green

18. Implement the following counter using JK FF:

Green->Yellow->Red->Yellow->Green

19. Implement the following counter using JK FF: 1->2->3->5->7->11->13->1

20. Design a 3 input NAND gate using 2 x 4 decoder(s). (Consider A,B,C as inputs where A is MSB and C is LSB)

21. Design a circuit that takes a 3 bit number as input subtracts 3 from it and shows the output.

Note: if the answer is less than 0 show 7 as output.

22.

Given the state diagram as follows, get the sequential circuit using SR flipflop.

